

THE "HETEROSIS" FINITE ELEMENT FOR PLATE BENDING

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Abstract—A new, high-accuracy, finite element for thick and thin plate bending is developed, based upon Mindlin plate theory. The element is a 9-node quadrilateral, which exhibits improved characteristics in comparison with the 8-node serendipity, or the 9-node Lagrange elements. In particular, the element stiffness possesses correct rank, and high accuracy is attainable for extremely thin plates. Due to the consistently good performance of the element, it is proposed as a candidate for inclusion in finite element programs available to the general user.

1. INTRODUCTION

Several recent studies [1-4] have exhibited the high accuracy attainable by isoparametric plate elements derived from Mindlin's theory [5]. It has also been pointed out that many of the problems which beset finite element applications of the classical thin plate theory are obviated by the Mindlin theory [1]. It is thus no wonder that an increased reliance on finite elements derived from Mindlin's theory, for both thick and thin plates, is currently taking place (see [1-4, 6-10] and references therein).

Nevertheless, it is the authors' view that none of the elements developed so far are suitable for inclusion in finite element codes which are widely disseminated. The reason for this is that the most important attribute of elements in the hands of the average user is *reliability*. Elements which occasionally "act up," or which require taking special precautions, are truly dangerous for general use. Although many of the elements derived from Mindlin's theory are highly accurate, simple to implement, efficient, and easy to use, none come without some admonition of potential difficulties. For example, the reduced integration, Lagrange quadrilaterals [1, 3, 4] are generally the most accurate Mindlin elements, yet *all* of those studied thus far possess at least one spurious zero-energy mode. In most applications, these are no problem, but occasionally they act up. Another example is the popular, 8-node serendipity element [11]. In the majority of applications, this element is capable of high accuracy, but bizarre convergence characteristics and divergence for very thin plates have on occasion been observed [3, 4].

It is the purpose of the present paper to describe a new plate finite element which possesses the accuracy and simplicity of the better Mindlin elements and, at the same time, has no shortcomings. The element is a 9-node quadrilateral which employs serendipity shape functions for the transverse displacement, w , and Lagrange shape functions for the rotations, θ_1, θ_2 . A selective, reduced integration scheme is used [1]. The new element possesses correct rank, is invariant, and passes the patch test appropriate to the Mindlin theory (see [1] for further discussion). It has performed consistently well on all of the numerical tests which we have subjected it to, including cases in which the serendipity and Lagrange

elements are poor. Pertinent information for the new element is presented in Fig. 1, along with descriptions of the related serendipity and Lagrange elements. We have named our new element the "heterosis element" since, in genetics, the improvement in characteristics exhibited by hybrids over those of the parents is called "heterosis."

The *constraint index* [12] is a heuristic measure of the ability of an element to accommodate the zero shear strain constraint, which is engendered in the thin-plate limit. For quadratic-level elements, a constraint index of 4 is considered optimal. A constraint index less than, or equal to, zero generally implies divergence in the thin plate limit. From Fig. 1, we see that the Lagrange element is optimal, the heterosis element is nearly optimal, and the serendipity element is poor. Our numerical experience has indicated a close correlation between constraint indices and the ability of Mindlin-type elements to successfully handle thin-plate problems.

We were lead to development of the heterosis element by the implications of some of the results obtained in [1]. In particular, we observed that the 9-node Lagrange element, employing the "S2" selective integration scheme, possesses only one spurious zero-energy mode. Furthermore, we noted that this mode would be eliminated by adopting the 8-node serendipity shape functions for the transverse displacement. The fact that the heterosis element possesses a favorable constraint index also suggested that it would be numerically successful in application to thin plates.

The outline of the remainder of the paper is given as follows:

The formulation of the element stiffness matrix and consistent load vector is briefly described in Section 2.

Numerical results for the three elements of Fig. 1 are presented in Section 3.

Our conclusions are contained in Section 4.

2. FINITE ELEMENT STIFFNESS MATRIX AND LOAD VECTOR

The general theory of Mindlin plates is discussed in [1]. Expressions for the finite element stiffness matrix and load vector are presented in [1] where it was assumed that the transverse displacement and rotations were defined in terms of identical shape functions, and identical nodal patterns. The heterosis element requires a relaxation of this assumption.

Let A^e and s^e denote the area and boundary, respectively, of a typical element. Let $N_1, N_2, \dots, N_8$ and $P_1,$

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	"serendipity"	"heterosis"	"Lagrange"
w shape functions	serendipity	serendipity	Lagrange
θ_1, θ_2 shape functions	serendipity	Lagrange	Lagrange
integration scheme	U2	S2	S2
number of spurious zero-energy modes	1*	0	1
constraint index	1	3	4

Key:

● w, θ_1, θ_2 degrees of freedom

○ θ_1, θ_2 degrees of freedom

U2 - 2 × 2 Gauss

S2 - {bending - 3 × 3 Gauss
shear - 2 × 2 Gauss

* Not communicable in a mesh of two or more elements

Fig. 1. Quadrilateral plate elements.

$P_2, \dots, P_9$ denote the serendipity, and Lagrange shape functions, respectively. (The center node is taken to be number nine. The ordering of the first eight nodes is arbitrary).

Following standard procedures, the element stiffness matrix, $\mathbf{k}^e$, is given by

$$\mathbf{k}^e = \mathbf{k}_b^e + \mathbf{k}_s^e \quad (1)$$

where

$$\mathbf{k}_b^e = \int_{A^e} \mathbf{B}^{bT} \mathbf{D}^b \mathbf{B}^b dA \quad \text{bending stiffness} \quad (2)$$

$$\mathbf{k}_s^e = \int_{A^e} \mathbf{B}^{sT} \mathbf{D}^s \mathbf{B}^s dA \quad \text{shear stiffness} \quad (3)$$

$$\mathbf{B}^b = [\mathbf{B}_1^b, \mathbf{B}_2^b, \dots, \mathbf{B}_9^b] \quad (4)$$

$$\mathbf{B}^s = [\mathbf{B}_1^s, \mathbf{B}_2^s, \dots, \mathbf{B}_9^s] \quad (5)$$

$$\mathbf{B}_a^b = \begin{bmatrix} 0 & P_{a,x} & 0 \\ 0 & 0 & P_{a,y} \\ 0 & P_{a,y} & P_{a,x} \end{bmatrix} \quad 1 \leq a \leq 8 \quad (6)$$

$$\mathbf{B}_a^s = \begin{bmatrix} N_{a,x} & -P_a & 0 \\ N_{a,y} & 0 & -P_a \end{bmatrix} \quad 1 \leq a \leq 8 \quad (7)$$

$$\mathbf{B}_9^b = \begin{bmatrix} P_{9,x} & 0 \\ 0 & P_{9,y} \\ P_{9,y} & P_{9,x} \end{bmatrix} \quad (8)$$

$$\mathbf{B}_9^s = \begin{bmatrix} -P_9 & 0 \\ 0 & -P_9 \end{bmatrix} \quad (9)$$

and $\mathbf{D}^b$ and $\mathbf{D}^s$ are matrices containing material coefficients and geometric data. For example, in the isotropic case

$$\mathbf{D}^b = \frac{t^3}{12} \begin{bmatrix} 2\mu + \bar{\lambda} & \bar{\lambda} & 0 \\ \text{symm.} & 2\mu + \bar{\lambda} & 0 \\ & & \mu \end{bmatrix} \quad (10)$$

and

$$\mathbf{D}^s = \kappa t \mu \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad (11)$$

where t is the thickness, $\bar{\lambda} = 2\lambda\mu/(\lambda + 2\mu)$, λ and μ are the Lamé parameters, and κ is a "shear correction factor," which is taken to be 5/6 throughout. In eqns (6)–(9) a comma denotes partial differentiation (e.g. $P_{a,x} = \partial P_a / \partial x$).

The element stresses may be obtained from the following relations:

$$\begin{Bmatrix} m_{xx} \\ m_{yy} \\ m_{xy} \end{Bmatrix} = -\mathbf{D}^b \mathbf{B}^b \mathbf{d}^e \quad \text{bending moments} \quad (12)$$

$$\begin{Bmatrix} q_x \\ q_y \end{Bmatrix} = \mathbf{D}^s \mathbf{B}^s \mathbf{d}^e \quad \text{shear resultants} \quad (13)$$

where

$$\mathbf{d}^e = \{d_i^e\} \quad (14)$$

$$d_i^e = \begin{cases} w_a, & I = 3a - 2, & 1 \leq a \leq 8 \\ \theta_{aa}, & \begin{cases} I = 3a + \alpha - 2, & 1 \leq a \leq 8, \\ I = 24 + \alpha, & a = 9, \end{cases} & \begin{matrix} \alpha = 1, 2 \\ \alpha = 1, 2 \end{matrix} \end{cases} \quad (15)$$

The element consistent load vector, $\mathbf{f}^e$, is given by

$$\mathbf{f}^e = \{f_i^e\} \quad (16)$$

$$f_i^e = \begin{cases} \int_{A^e} N_a F dA + \int_{s^e \cap s_2} N_a Q ds, & I = 3a - 2, \quad 1 \leq a \leq 8 \\ \int_{A^e} P_a C_a dA + \int_{s^e \cap s_2} P_a M_a ds \end{cases} \quad \text{shape functions, i.e.} \quad (17)$$

$$x_9 = \sum_{a=1}^8 N_a(0,0)x_a \quad (18)$$

$$y_9 = \sum_{a=1}^8 N_a(0,0)y_a \quad (19)$$

$$\begin{cases} I = 3a + \alpha - 2, & 1 \leq a \leq 8, & \alpha = 1, 2 \\ I = 24 + \alpha, & a = 9, & \alpha = 1, 2 \end{cases}$$

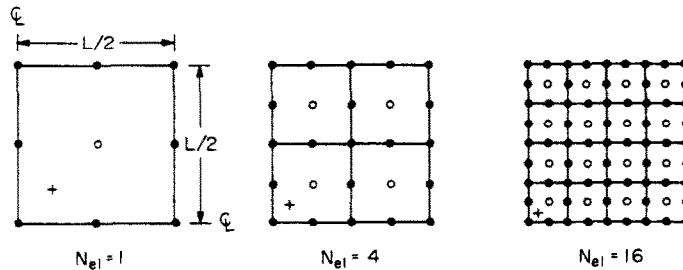
where F is the total applied transverse force per unit area, C_a is the total applied couple per unit area, Q is the applied boundary shear force, M_a is the applied boundary moment, and s_2 is the portion of the plate boundary upon which forces and moments are prescribed.

To avoid any confusion in the geometric definition of the element, we stipulate that the global Cartesian coordinates of the center node, x_9, y_9 , are located at the origin of the isoparametric coordinates of the serendipity

3. NUMERICAL EXAMPLES

All computations were run on an IBM 370/158 computer in double precision (64 bits per floating-point word). A Poisson's ratio of 0.3 and a Young's modulus of 10.92×10^5 were used throughout.

Two types of simply-supported boundary conditions are used in the calculations: SS_1 , in which only the transverse displacement is set to zero, and SS_2 , in which the transverse displacement and tangential rotation are set to zero. In applications to thin plates, SS_1 is preferable since it leads to convergent results when polygonal approximations of curved boundaries are employed,



+ denotes location of Gauss point (2x2 rule) nearest center
Fig. 2. Square plate. Due to symmetry, only one quadrant is discretized.

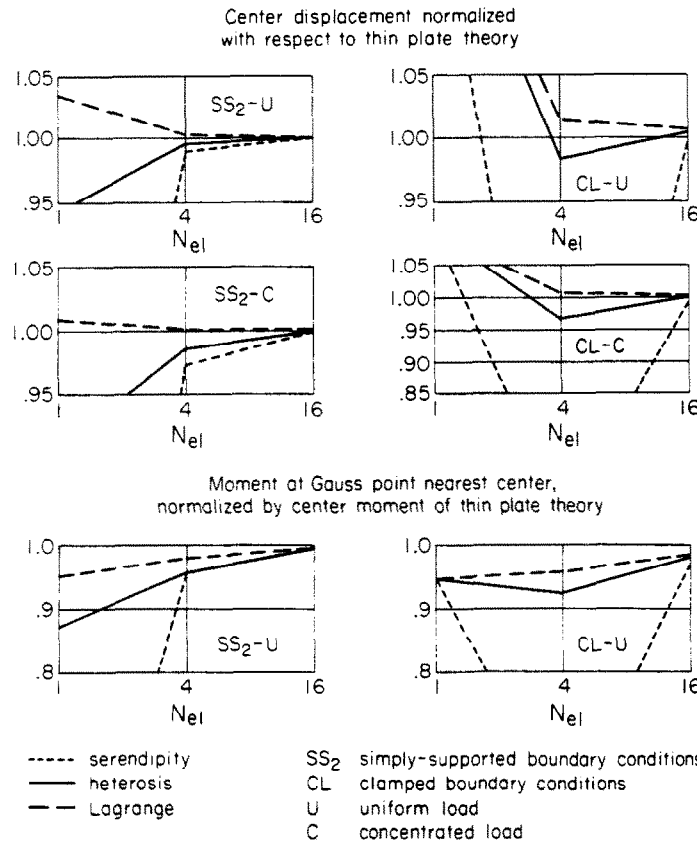


Fig. 3. Convergence study for thin square plate.

whereas SS_2 generally does not. For elaboration on these boundary-condition "paradoxes," the reader is referred to [1].

3.1 Square plate

Mesh patterns are depicted in Fig. 2. The geometric data employed where $L = 10$ and t (thickness) $= 0.1$.

The results of a convergence study are presented in Fig. 3. The serendipity element exhibits particularly poor convergence characteristics for the clamped cases. The Lagrange and heterosis elements demonstrate good convergence properties, with the former being somewhat more accurate.

The ability of the elements to handle extremely thin plates is illustrated in Fig. 4. Both the Lagrange and heterosis elements exhibit "stable" convergence characteristics for large length-to-thickness ratios. Numerical degeneration commences at $L/t = 10^6$ because the shear stiffness is $O(L^2/t^2)$ times the bending stiffness, resulting in the bending effects being overwhelmed by roundoff errors. The "plateaus" exhibited by the Lagrange and heterosis elements, up to $L/t = 10^6$, may be extended *ad infinitum* by the technique proposed in [2]. On the other hand, the serendipity element diverges rapidly for relatively small values of L/t (i.e. $O(10^2)$).

3.2 Circular plate

The meshes for this problem are depicted in Fig. 5. A value of $R = 5$ is used for all cases. Thin plate (i.e. $t = 0.1$) convergence results are presented in Fig. 6. The heterosis element is consistently superior to the serendipity element.

As we have seen so far, the presence of the spurious zero-energy mode generally does not adversely affect the accuracy of the Lagrange element. The reason for this is that when two adjacent nodal values of transverse displacement are set to zero, the mode is no longer present (i.e. the global stiffness matrix is not rank deficient). However, when the element thickness-to-length ratio increases beyond a certain value, the mode is only weakly coupled to the boundary, and may become manifest in certain singular situations. This is illustrated in Fig. 7, where significant oscillations in transverse displacement may be noted. (In Fig. 7 r denotes the radial coordinate, D is the flexural rigidity and P is the amplitude of the uniform load). The heterosis element, having correct rank, does not exhibit this deficiency. (For this problem, the serendipity results are almost identical to the heterosis results. At points at which the plot symbols tended to overlap, we included only the heterosis symbol).

4. CONCLUSIONS

The object of the research reported upon herein was to develop a high-accuracy, finite element for thick and thin plate bending, which had no ostensible fallibility. We believe that the heterosis element achieves this objective. It has consistently performed well on all of the numerical tests to which we have subjected it, and, in particular, it has exhibited good behavior in cases which have manifested weaknesses of other plate elements. In the hands of knowledgeable and experienced analysts, many different plate elements can be used effectively, even those which occasionally "act up." However, for

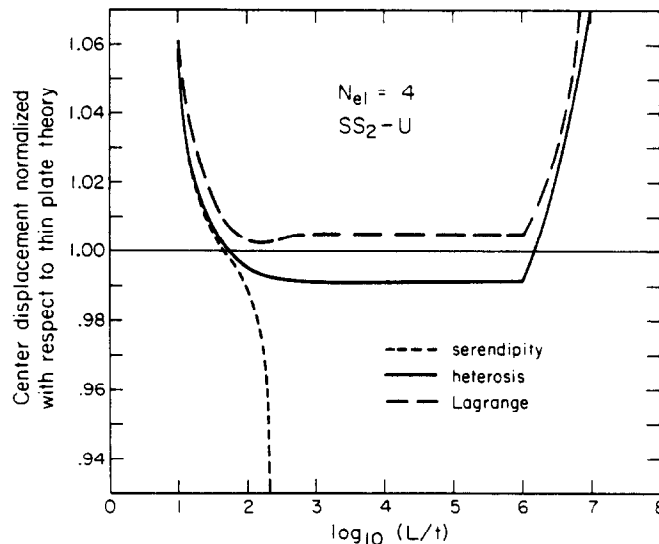
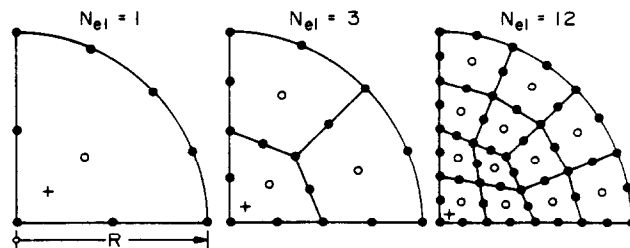


Fig. 4. Aspect ratio study for square plate.



+ denotes location of Gauss point (2×2 rule) nearest center
Fig. 5. Circular plate. Due to symmetry, only one quadrant is discretized.

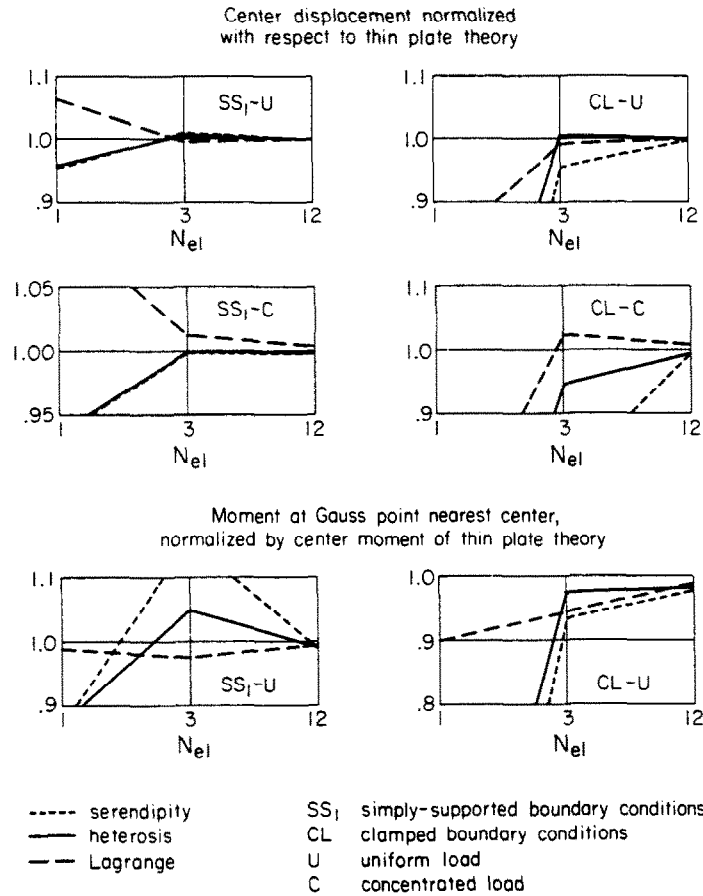


Fig. 6. Convergence study for thin circular plate.

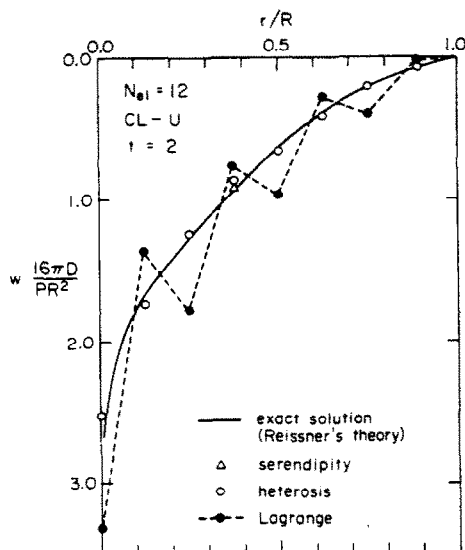


Fig. 7. Comparison of transverse displacements for thick circular plate.

the average user of finite element programs, more fool-proof elements are desirable. The reliability and simplicity of the heterosis element make it an obvious candidate for inclusion in finite element programs available to the average user.

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